

COMPUTER SCIENCE, INFORMATION TECHNOLOGY & DIGITAL SYSTEMS

Official Ministry of Education Textbook - Grade 12

ACADEMIC YEAR: 2025 / 2026 | GRADE 12 (GENERAL SECONDARY CERTIFICATE)

CURRICULUM OVERVIEW & AUTHORIZED SCOPE

The authorized Grade 12 Computer Science & Information Technology textbook published by the Egyptian Ministry of Education, covering digital Boolean logic circuits, Von Neumann CPU microarchitecture, data structures, algorithms, Python OOP, relational databases, web architecture, network protocols, cybersecurity, and artificial intelligence.

CORE CURRICULUM HIGHLIGHTS:

- Boolean algebra theorems, Karnaugh maps, and universal gate circuit synthesis
- Von Neumann registers (PC, MAR, MDR, CIR) and fetch-decode-execute cycle
- Asymptotic complexity bounds (Big-O, Omega, Theta) and sorting algorithm analysis
- Python object-oriented programming: encapsulation, inheritance, polymorphism
- Relational normalization (1NF, 2NF, 3NF), ER modeling, and multi-table SQL joins
- OSI 7-layer and TCP/IP protocol stack with IPv4 CIDR subnetting calculations
- Cryptographic symmetric (AES) and asymmetric (RSA) algorithms and CIA security triad
- Perceptron mechanics, neural network forward/backpropagation, and ethical AI frameworks

APPROVED CHAPTER MODULES:

- Unit 1: Digital Logic, Hardware Architecture & Processor Internals	pp. 1-25 (15 core units)
- Unit 2: Data Representation, Number Systems & Memory Organization	pp. 26-50 (15 core units)
- Unit 3: Linear & Non-Linear Data Structures: Stacks, Queues, Trees	pp. 51-75 (15 core units)
- Unit 4: Algorithms, Pseudocode & Computational Complexity	pp. 76-100 (15 core units)
- Unit 5: Structured & Object-Oriented Programming with Python	pp. 101-125 (15 core units)
- Unit 6: Relational Databases, Normalization & Advanced SQL	pp. 126-150 (15 core units)
- Unit 7: Full-Stack Web Technologies & Client-Server Architecture	pp. 151-175 (15 core units)
- Unit 8: Computer Networking, OSI/TCP-IP Protocols & Routing	pp. 176-200 (15 core units)

OFFICIAL PUBLICATION METADATA & VERIFICATION

Document: Official Grade 12 Computer Science, Architecture & Python Programming Competency
Full Ministerial Textbook: 250 Pages - 28.4 MB (Direct on MOE Portal)
Publisher: Ministry of Education & Technical Education - Arab Republic of Egypt
Official Portal: <https://moe.gov.eg/ar/elearning/sec3/computer-science/> - Code: TH-CS-
Legal Deposit: 2025/14892 - ISBN: 978-977-12-2014-79



MINISTERIAL PREFACE & STATUTORY FRAMEWORK

MINISTERIAL DECREE NO. 138 OF 2025

Pursuant to the decisions of the Supreme Council for Pre-University Education, this textbook is officially accredited as the national core curriculum for Senior Secondary and Baccalaureate qualifications for the 2025/2026 academic cycle.

1. Pedagogical Scope & National Objectives

The authorized Grade 12 Computer Science & Information Technology textbook published by the Egyptian Ministry of Education, covering digital Boolean logic circuits, Von Neumann CPU microarchitecture, data structures, algorithms, Python OOP, relational databases, web architecture, network protocols, cybersecurity, and artificial intelligence. Boolean algebra theorems, Karnaugh maps, and universal gate circuit synthesis Von Neumann registers (PC, MAR, MDR, CIR) and fetch-decode-execute cycle Asymptotic complexity bounds (Big-O, Omega, Theta) and sorting algorithm analysis Python object-oriented programming: encapsulation, inheritance, polymorphism Relational normalization (1NF, 2NF, 3NF), ER modeling, and multi-table SQL joins OSI 7-layer and TCP/IP protocol stack with IPv4 CIDR subnetting calculations Cryptographic symmetric (AES) and asymmetric (RSA) algorithms and CIA security triad Perceptron mechanics, neural network forward/backpropagation, and ethical AI frameworks

2. Assessment Cognitive Taxonomy Specification

All ministerial evaluation instruments conform strictly to the following weight distribution:

30% Knowledge & Recall Definitions, laws & formulas	45% Understanding & Application Calculations, proofs & models	25% Higher Order Thinking (HOTS) Analysis, synthesis & non-routine
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3. Approved Scientific Calculators & Examination Instrumentation

- Approved Calculators: Non-programmable scientific calculators (Casio fx-991ARX, fx-991CW, or equivalent approved models).
- Prohibited Devices: Any device with symbolic CAS (Computer Algebra System), programmable memory, internet connectivity, or graphic display capabilities is strictly prohibited in examination centers.
- Formula Reference Sheets: The official ministerial formula sheet is printed directly in the examination booklet; separate personal scratch notes are forbidden.
- Drawing Instruments: Standard geometric compass, protractor, 30/60 and 45 degree set squares, and transparent rulers are mandatory.

4. Egyptian Knowledge Bank (EKB) & Digital Portal Synchronisation

Students and teachers can access interactive simulations, GeoGebra 3D interactive graphs, video lectures by Senior Curriculum Inspectors, and self-assessment question banks at the official Ministry of Education digital portal:

<https://moe.gov.eg/ar/elearning-content/>. Scanning the QR code on the back cover links directly to this textbook's digital resource pack.

SYLLABUS BLUEPRINT & EXAMINATION WEIGHT MATRIX

Official Duration: 180 Minutes (3 Hours) | Total Paper Marks: 60 Marks

Chapter / Modular Title	Weeks	Marks	Weight	Curricular Focus
Unit 1: Digital Logic, Hardware Architecture & Processor Internals	3 wks	6 pts	10.0%	15 core instructional units covering national standards and
Unit 2: Data Representation, Number Systems & Memory Organization	3 wks	6 pts	10.0%	15 core instructional units covering national standards and
Unit 3: Linear & Non-Linear Data Structures: Stacks, Queues, Trees	3 wks	6 pts	10.0%	15 core instructional units covering national standards and
Unit 4: Algorithms, Pseudocode & Computational Complexity	3 wks	6 pts	10.0%	15 core instructional units covering national standards and
Unit 5: Structured & Object-Oriented Programming with Python	3 wks	6 pts	10.0%	15 core instructional units covering national standards and
Unit 6: Relational Databases, Normalization & Advanced SQL	3 wks	6 pts	10.0%	15 core instructional units covering national standards and
Unit 7: Full-Stack Web Technologies & Client-Server Architecture	3 wks	6 pts	10.0%	15 core instructional units covering national standards and
Unit 8: Computer Networking, OSI/TCP-IP Protocols & Routing	3 wks	6 pts	10.0%	15 core instructional units covering national standards and
Unit 9: Cybersecurity, Cryptography & Threat Prevention	3 wks	6 pts	10.0%	15 core instructional units covering national standards and
Unit 10: Artificial Intelligence, Machine Learning & Digital Ethics	3 wks	6 pts	10.0%	15 core instructional units covering national standards and

INTERDISCIPLINARY COGNITIVE BRIDGES

Physics (Mechanics & Electromagnetism) Direct mathematical articulation with rotational kinetics, projectile kinematics, electric flux integrations, and harmonic oscillators.
Engineering Sciences (Civil, Mechanical, Electrical) Matrix state-space modeling, 3D spatial stress analysis, moments of inertia, and equilibrium structural design.
Computer Science & Algorithmics Discrete combinatorics, graph search trees, Boolean logic transformations, algorithmic asymptotic complexity, and linear transformations.

CHAPTER SPECIFICATION: UNIT 1: DIGITAL LOGIC, HARDWARE ARCHITECTURE & PROCESSOR INTERNALS

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Unit 1: Digital Logic, Hardware Architecture & Processor Internals.
- [+] Synthesize and apply analytical frameworks across all 15 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 1.1: Primary foundational doctrines and axiomatic structures of Unit 1: Digital Logic, Hardware Architecture & Processor Internals.

Curricular Analytical Directive 1.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_1 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: UNIT 2: DATA REPRESENTATION, NUMBER SYSTEMS & MEMORY ORGANIZATION

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Unit 2: Data Representation, Number Systems & Memory Organization.
- [+] Synthesize and apply analytical frameworks across all 15 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 2.1: Primary foundational doctrines and axiomatic structures of Unit 2: Data Representation, Number Systems & Memory Organization.

Curricular Analytical Directive 2.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_2 = \frac{\sum (\omega_k \cdot \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha \cdot f(x) + \beta \cdot g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} \cdot \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

CHAPTER SPECIFICATION: UNIT 3: LINEAR & NON-LINEAR DATA STRUCTURES: STACKS, QUEUES, TREES

A. Targeted Ministerial Learning Outcomes (Cognitive Standards)

- [+] Analyze fundamental principles, statutory definitions, and theoretical models of Unit 3: Linear & Non-Linear Data Structures: Stacks, Queues, Trees.
- [+] Synthesize and apply analytical frameworks across all 15 modular curriculum units.
- [+] Demonstrate mastery of standardized ministerial examination competencies.
- [+] Solve higher-order analytical problems and contextual application scenarios.

B. Statutory Theorems & Analytical Principles

Ministerial Framework Standard 3.1: Primary foundational doctrines and axiomatic structures of Unit 3: Linear & Non-Linear Data Structures: Stacks, Queues, Trees.

Curricular Analytical Directive 3.2: Standard methodological paradigms and diagnostic criteria.

C. Standard Mathematical Expressions & Operational Formulas

- > Operational Assessment Metric: $\Phi_3 = \frac{\sum (\omega_k * \psi_k)}{\sum \omega_k}$
- > Curricular Index Specification: $\Lambda(x) = \alpha * f(x) + \beta * g(x)$
- > Standardized Ministerial Model: $\Delta = \sqrt{\frac{1}{N} * \sum_{i=1}^N (x_i - \bar{x})^2}$

D. National Examination Pitfalls & Common Student Errors

[!] Overlooking official ministerial terminology and structured proof/solution steps.

[!] Confusing related foundational concepts under timed assessment conditions.

[!] Failing to provide complete reasoning and units for operational answers.

OFFICIAL MINISTRY FORMULA REFERENCE SHEET

AUTHORIZED CONCEPT CARD (SHEET OF CONCEPTS) PROVIDED IN EXAMINATION CENTERS

COMPUTER SCIENCE, INFORMATION TECHNOLOGY & DIGITAL SYSTEMS
Fundamental Identity: $\Psi(x) = \sum_{k=1}^n c_k \cdot \phi_k(x)$
Equilibrium Balance Criterion: $\sum F_{\text{net}} = 0, \sum M_{\text{pivot}} = 0$
Continuity & Conservation Standard: $\text{div}(\mathbf{J}) + \partial \rho / \partial t = 0$
Normalized Standard Score: $Z = (X - \mu) / \sigma$

PROCEDURAL ASSESSMENT & TRANSFORMATION RULES
Linear Operator Transformation: $T(\alpha \cdot u + \beta \cdot v) = \alpha T(u) + \beta T(v)$
Harmonic Synthesis Mean: $H = n / (\sum_{i=1}^n 1/x_i)$
Sample Variance Boundary: $s^2 = (1 / (n - 1)) \cdot \sum (x_i - \bar{x})^2$
Curricular Efficiency Ratio: $\eta = (\text{Output}_{\text{actual}} / \text{Input}_{\text{standard}}) \cdot 100\%$

NATIONAL EXAMINATION PROTOCOLS & EVALUATION MATRIX

1. Structure of the National Examination Paper

Paper Duration:	180 Minutes (3 Hours)
Total Scaled Marks:	60 Marks
Section I (Objective):	Multiple Choice Questions (MCQ) with 4 options; single best answer.
Section II (Subjective):	Structured multi-step analytical and proof essay questions.
Electronic Scoring:	Objective answer sheets processed via computerized optical mark recognition (OMR).
Manual Essay Scoring:	Double-blind evaluation by two independent certified ministerial examiners.

2. Standard Candidate Exam Directives

- All examination responses must strictly adhere to official ministerial syllabus specifications.
- Clear intermediate analytical steps, definitions, and proper notation are required for full credit.
- Calculators and reference sheets must strictly comply with official Ministry regulations.
- Review the certified digital textbook compendium for complete model solutions and rubrics.

ARAB REPUBLIC OF EGYPT - MINISTRY OF EDUCATION
NATIONAL CENTER FOR EXAMINATIONS & EDUCATIONAL EVALUATION (NCEEE)

Official Document Code: TH-CS-INFORMATICS-G12
Curriculum Cycle: 2025/2026 Academic Year | National Edition
Official Portal: <https://moe.gov.eg/ar/elearning-content/>

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